

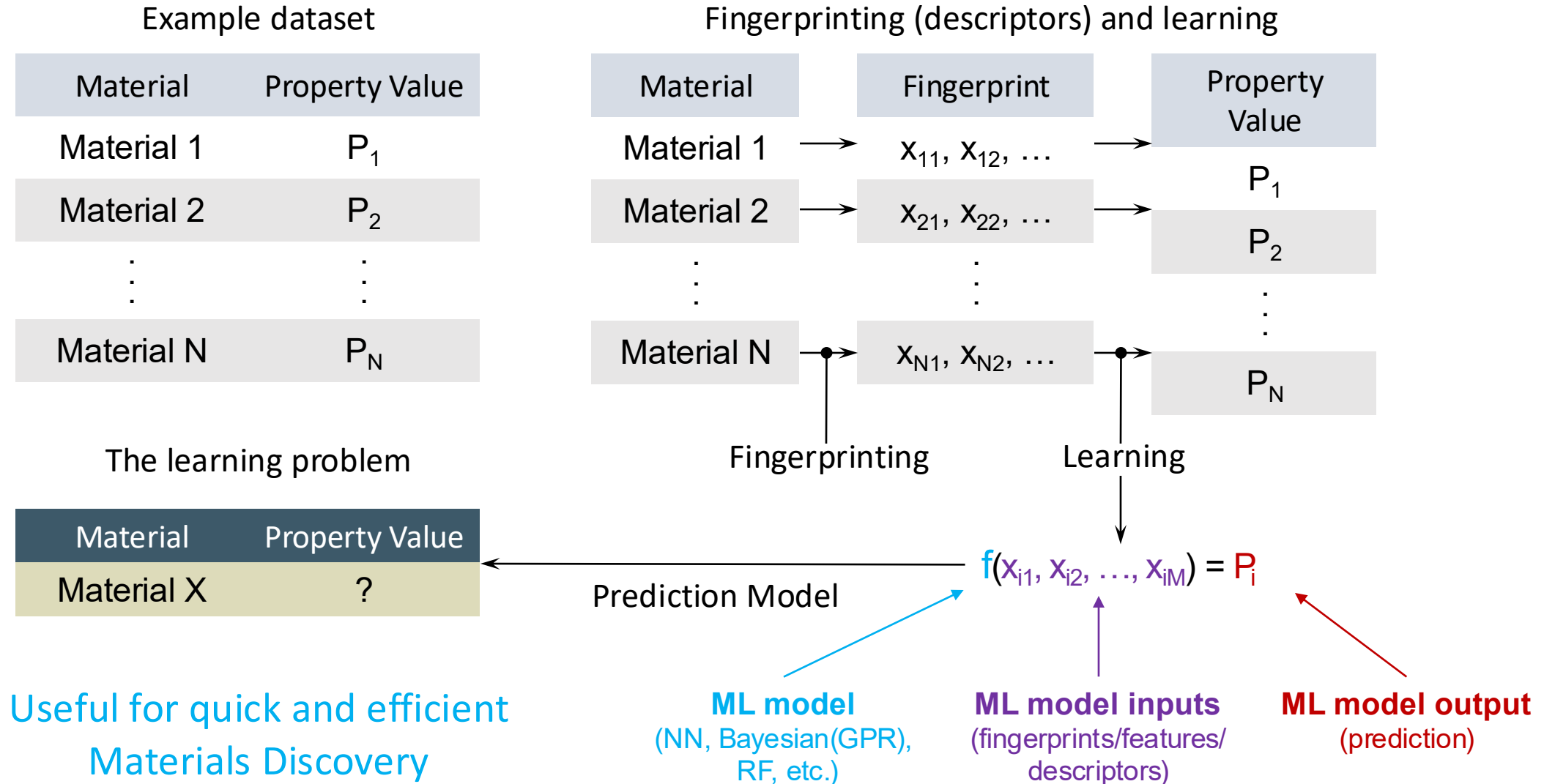
Polymer Informatics

Rohit Batra

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- Huge range of polymer fingerprints:
 - Molecular fragment level (e.g., count of C-C-C bonds)
 - Block level
 - Graph-based (chi)
- In some cases, fingerprints are learned by the ML model “directly”
 - Transformer-based models
 - GNN-based models

Machine Learning in Materials Science

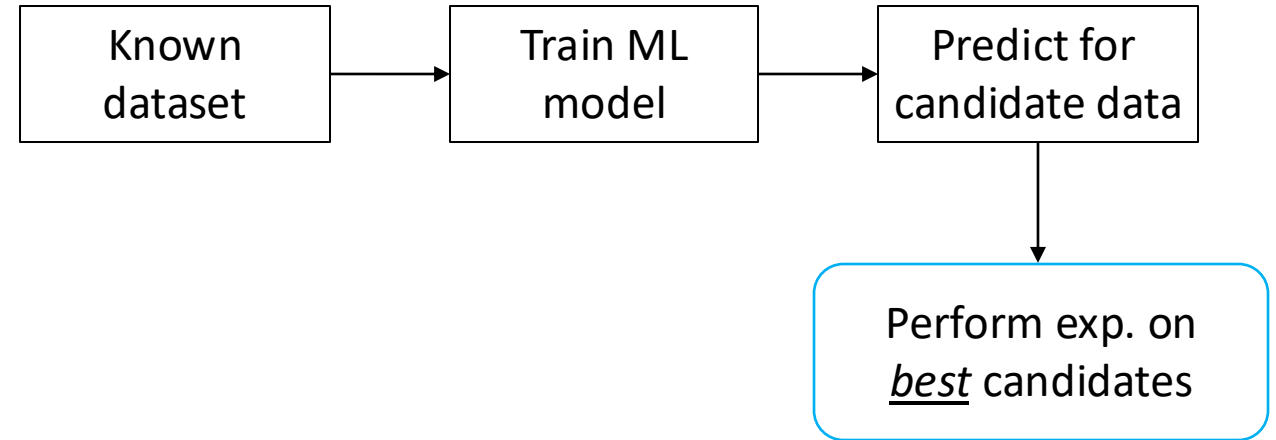


A Typical Materials Discovery Problem

Known dataset		Candidate dataset
Material	Property Value	Candidate Material
Material 1	P_1	Material 1
Material 2	P_2	Material 2
$\vdots$	$\vdots$	$\cdot$
Material N	P_N	$\cdot$
		$\cdot$
		$\cdot$
		$\cdot$
		$\cdot$
		$\cdot$
		$\cdot$
		Material Z

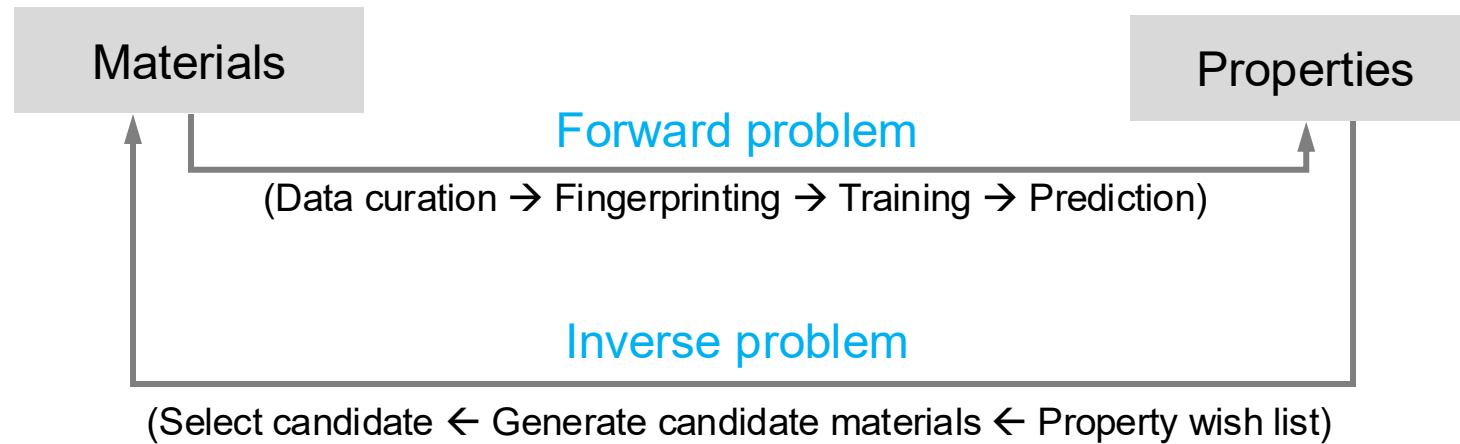
$N = 20-30$
 $Z = 1000 - 10^5$

(One) ML solution:



Many candidates need to be evaluated
– infeasible; limited budget

Materials (or Polymer) Design



ML Methods can help us quickly screen candidates with desired properties

Materials Fingerprint



Alloy	Hardness (HV)
$\text{Al}_5\text{Cr}_2\text{Cu}_2\text{Fe}_2\text{Ni}_4$	593
$\text{Al}_1\text{Cr}_2\text{Cu}_2\text{Fe}_2\text{Ni}_2$	382
...	...

SEM Images	Young's modulus (N/m ²)
Image 1	30
Image 2	96
...	...

Polymer	Li-ion conductivity (S/cm)
Polymer 1	7.2×10^{-4}
Polymer 2	2.5×10^{-2}
...	...

Exp. process	Structure
Anneal T_1 , cooling R_1	Image 1
Anneal T_2 , cooling R_2	Image 2
...	...

How to represent alloys, polymers, images and experimental process as numerical vectors?

Use material conc., atomic bonds, elemental properties (mass), pixel information, etc.

Fingerprint Considerations



A materials fingerprint:

- numerically represent materials or experimental process
- large dimensional vector

A GOOD materials fingerprint:

- easy to compute for new candidates
- uniquely represent candidate materials
- simple or non-complex (better learning)
- capture invariance
- depends on the target property
- requires domain expertise
- can even be learned using advanced ML

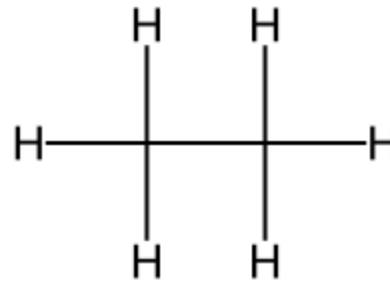
Fingerprint Construction Guidelines

- Should impact the material property in some manner (domain knowledge)
- Carefully think about the **mechanism** behind the target property
- Ensure that fingerprint vector uniquely represent train and test data
- The fingerprint vector should be easy to obtain for “new” cases
- Important to accept the limitations of the model

Representing molecular/polymer structure

In human readable form:

C₂H₆ – Ethane



In machine readable form:

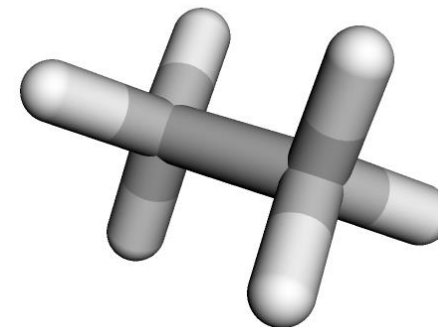
1. A robust way of representing - for the computers to read
2. Introduction to SMILES – (C₂H₆ -> CC)
3. Ever evolving area – new representations – eg: SELFIES

What is SMILES?

- SMILES (Simplified Molecular Input Line Entry System) chemical language
- SMILES has five basic syntax rules
- https://web.archive.org/web/20080328080430/https://www.epa.gov/med/Prods_Pubs/smiles.htm
- <https://www.sciencedirect.com/science/article/pii/S2666389922002069>

Rule One: Atoms and Bonds

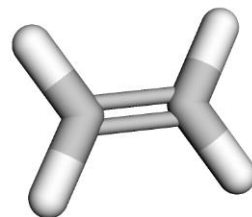
- SMILES supports all elements in the periodic table. An atom is represented using its respective atomic symbol.
- Upper case letters refer to non-aromatic atoms; lower case letters refer to aromatic atoms.
- If the atomic symbol has more than one letter the second letter must be lower case.
- Bonds are denoted as shown below:
 - - Single bond
 - = Double bond
 - # Triple bond
 - * Aromatic bond
 - . Disconnected structures
- Single bonds are the default and therefore need not be entered. For example, 'CC' means ethane molecule
- H atoms are implicitly assumed to complete valency
- A blank terminates the SMILES string.



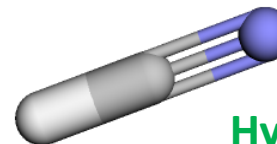
Rule Two: Simple Chains

- Chain structures are represented by combining atomic symbols and bond symbols
- Some examples:

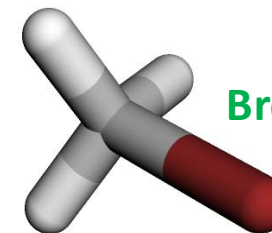
CC	CH ₃ CH ₃	Ethane
C=C	CH ₂ CH ₂	Ethene
CBr	CH ₃ Br	Bromomethane
C#N	C=N	Hydrocyanic acid
Na.Cl	NaCl	Sodium chloride



Ethene



Hydrocyanic acid



Bromomethane

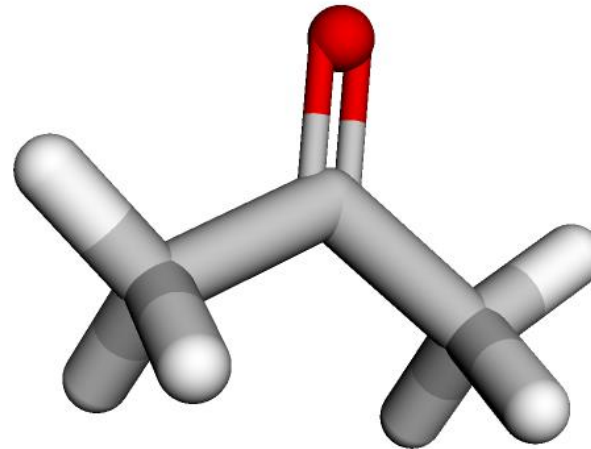
- H atoms can be explicitly included, but if one hydrogen bond is identified in the string, then all hydrogens should be included

HC(H)=C(H)(H) Ethene

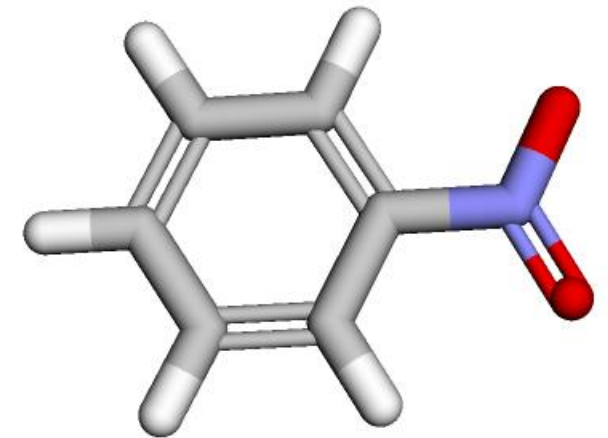
- Caution: Misinterpretations are possible. For example, 'Sc' could be interpreted as a sulfur atom connected to an aromatic carbon by a single bond, or it could be the symbol for scandium. There is a priority list to these interpretations. To identify scandium the user should enter [Sc].

Rule Three: Branches

- A branch from a chain is specified using parenthesis.
- If it is connected by a double or triple bond, the bond symbol immediately follows the left parenthesis.
- Some examples:
 - CC(O)C 2-Propanol
 - CC(=O)C 2-Propanone
 - CC(CC)C 2-Methylbutane
 - CC(C)CC(=O) 2-Methylbutanal
 - c1c(N(=O)=O)cccc1 Nitrobenzene
 - CC(C)(C)CC 2,2-Dimethylbutane



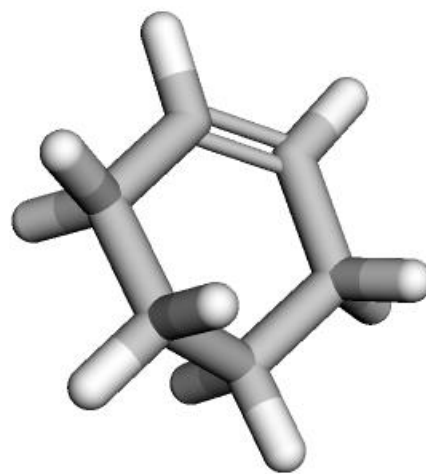
2-Propanone



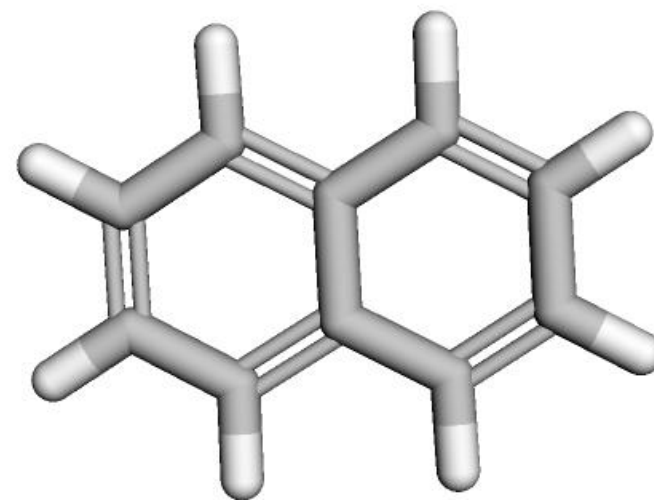
Nitrobenzene

Rule Four: Rings

- Opening and closing ring atoms identified using numbers
- For example, in C1CCCCC1, the first carbon has a number '1' which connects by a single bond with the last carbon which also has a number '1'. The resulting structure is cyclohexane.
- Multiple rings identified using different numbers for each ring
- If a double, single, or aromatic bond is used for the ring closure, the bond symbol is placed before the ring closure number.
- Some examples:
 - C=1CCCCC1 Cyclohexene
 - c1ccccc1 Benzene
 - C1OC1CC Ethyloxirane
 - c1cc2ccccc2cc1 Naphthalene



Cyclohexene



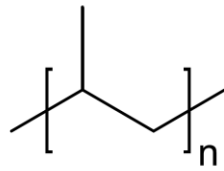
Naphthalene

Rule Five: Charged Atoms

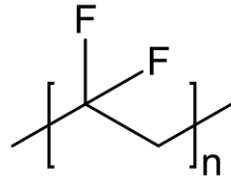
- Charges on an atom can be used to override the knowledge regarding valence completion
- Charged atom identified by atom symbol followed by brackets which enclose the charge on the atom. The number of charges may be explicitly stated (`{-1}`) or not (`{-}`).
- For example:
 - CCC(=O)O{-1} Ionized form of propanoic acid
 - or
 - CCC(=O)O{-}
 - c1ccccn{+1}1CC(=O)O 1-Carboxylmethyl pyridinium

A few Important Polymer concepts

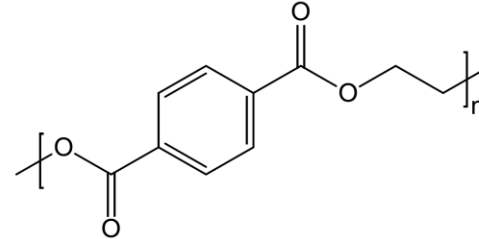
Polymers can be represented using SMILES *language*



[*]CC([*])C



[*]CC([*])(F)F



[*]CCOC(=O)c1ccc(C(=O)O[*])cc1

Polymers SMILES are not unique

1. Translation $(-OCC-)\xleftrightarrow{\text{translation}}(-COC-)\xleftrightarrow{\text{translation}}(-CCO-)$ [*]OCC[*], [*]COC[*], [*]CCO[*]

2. Addition/subtraction $(-COCO-)\xrightleftharpoons[\text{subtraction}]{\text{addition}}(-COCOCO-)$ [*]COCO[*], [*]COCOCO[*]

3. Multiple SMILES representation [*]CC([*])(F)F, [*]CC(F)(F)[*], [*]CC(F)([*])F

ML models should be invariant to these transformations

Fingerprint Considerations



A materials fingerprint:

- numerically represent materials or experimental process
- large dimensional vector

A GOOD materials fingerprint:

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- uniquely represent candidate materials
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- capture invariance
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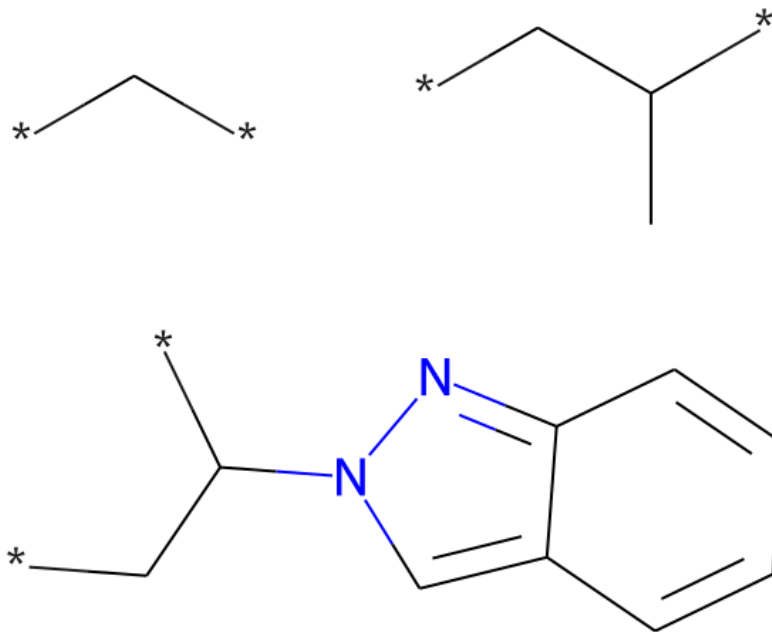
Fingerprint Construction Guidelines

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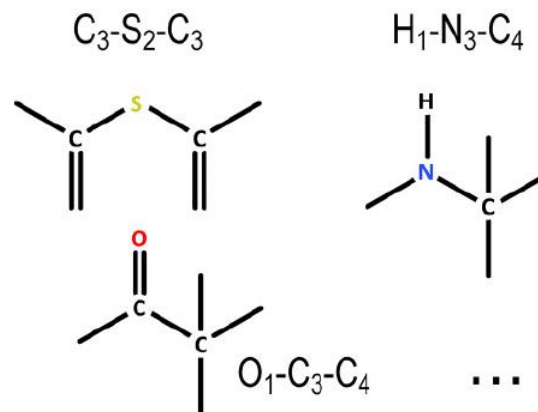
Let's develop fingerprints...

Polymer database:

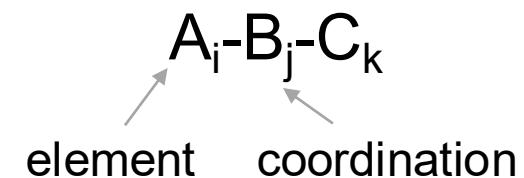
Polymer	E_{gc} (eV)
<chem>[*]C[*]</chem>	6.89
<chem>[*]CC([*])C</chem>	6.52
...	...
...	...
<chem>[*]CC([*])n1cc2ccccc2n1</chem>	3.76



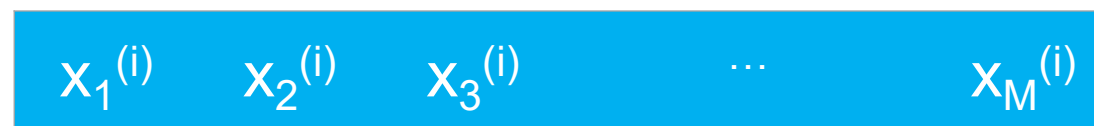
Molecular Fragment Level Fingerprints



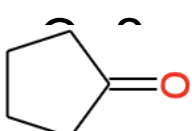
General formula of atom triples



J. Appl. Phys. 128, 171104 (2020)

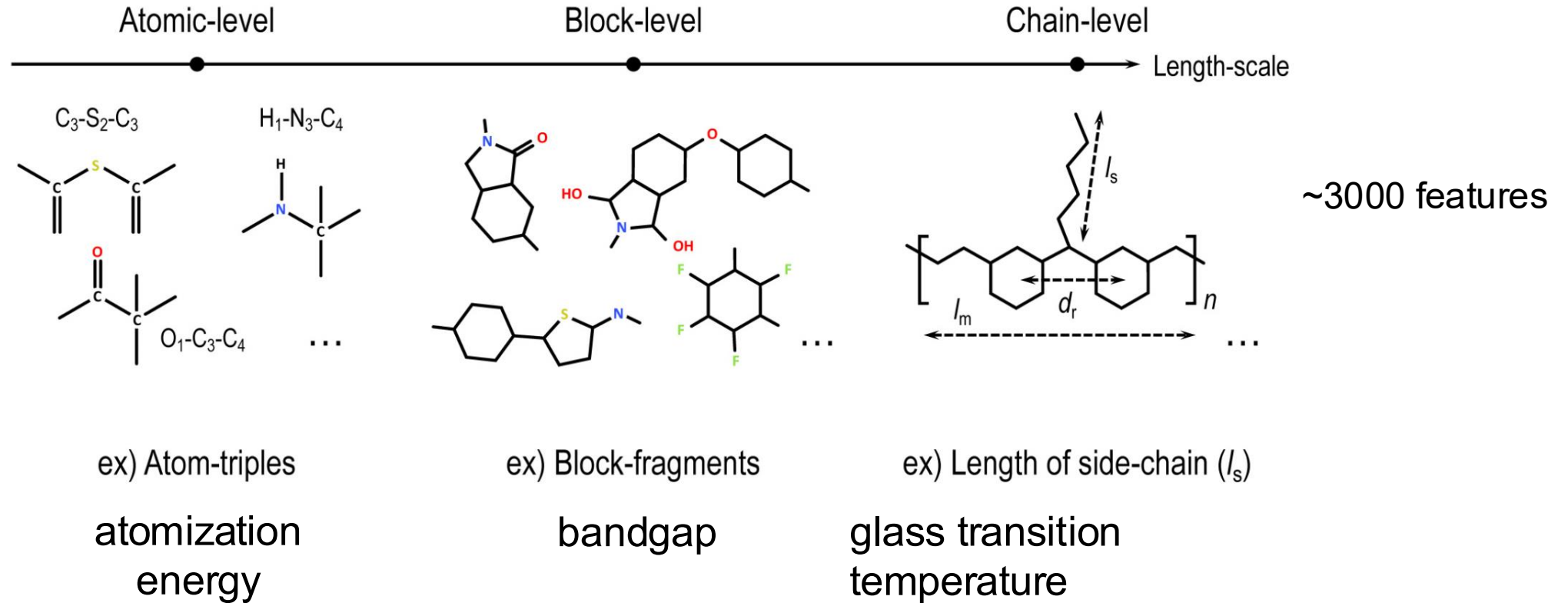


Fraction count of fragment 1

- Fraction of different fragments in the molecule/monomer units
- Total number of possible fragments are in hundreds (balance between sparsity and information)
- Is $O_1-C_3-C_4$ same as C_4-C
- What about fingerprint of 

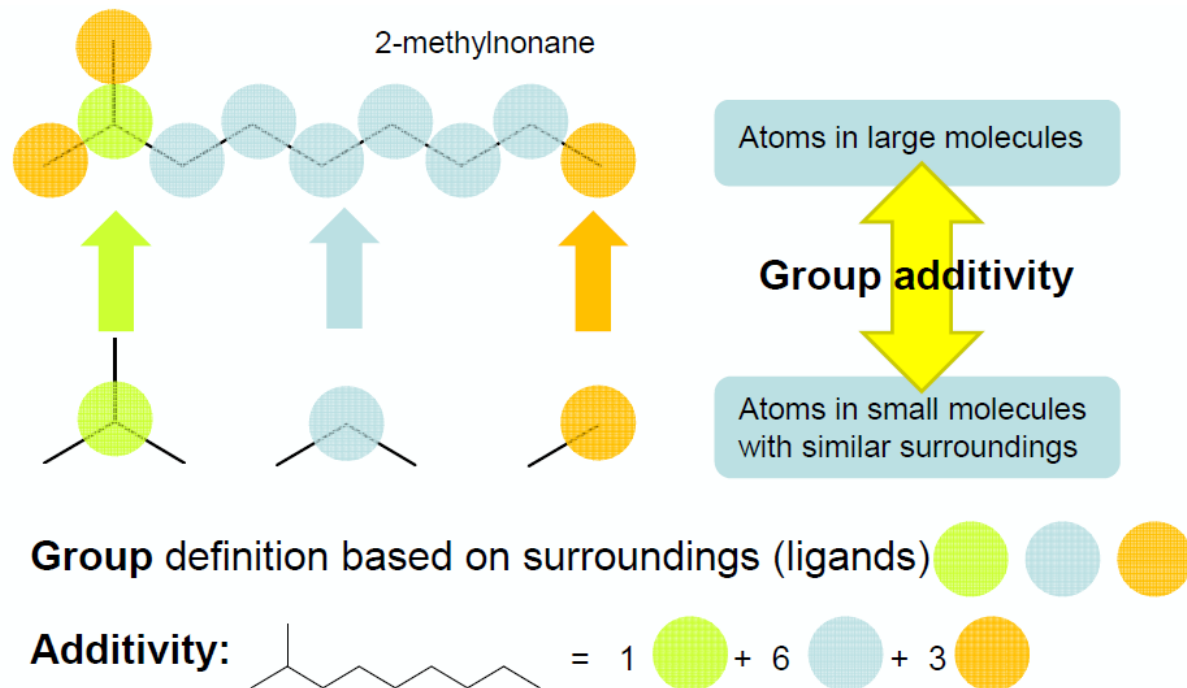
Molecular Fragment Level Fingerprints

J. Appl. Phys. 128, 171104 (2020)



- Different properties involve chemical phenomena at different length scales
- Overcome limitations of Group contribution method
- Unfortunately, fingerprint library is not open source!

Group contribution method (Traditional)



<https://github.com/ReactionMechanismGenerator/RMG-Py>

Krevelen, Nijenhuis. *Properties of Polymers: Their Correlation with Chemical Structure; their Numerical Estimation and Prediction from Additive Group Contributions* (Elsevier, 2009).

Limitations?

- Ad hoc definition of groups
- Linear dependence
- Same contribution of a group irrespective of other neighbouring groups

Success Story of Molecular-level fingerprints

Polymer Genome
An informatics platform for polymer property prediction and design using machine learning

Home Guide References Profile Sign out

Draw Polymer Predict Properties

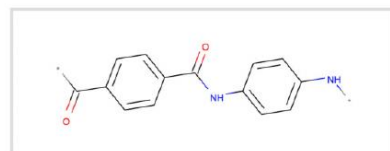
Polymer Name Kevlar poly-p-phenyleneterephthalamide
Poly(imino-1,4-phenyleneimino-terephthaloyl) PPPT

SMILES *[N]c1ccc(NC(=O)c2ccc(C([*])=O)cc2)cc1 Polyamides/thioamides

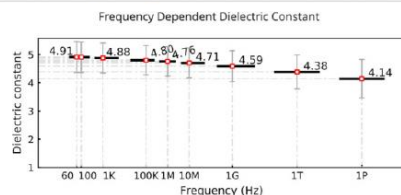
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Electronic Properties

Bandgap (Bulk)	2.9 ± 0.5 eV
Bandgap (Single Chain)	3.2 ± 0.4 eV
Ionization Energy	6.2 ± 0.4 eV
Electron Affinity	2.1 ± 0.4 eV



Dielectric & Optical Properties



Dielectric Constant (Crystal)	4.6 ± 0.5
Refractive Index (Bulk Resin)	1.73 ± 0.07
Refractive Index (Crystal)	1.9 ± 0.1

Mechanical Properties

Tensile Strength	69.7 ± 41.6 MPa
Young's Modulus	9.29 ± 1.24 GPa

Solubility Properties

Solvent NMP DMAc

Non-solvent 1,4-dioxane Chloroform Ethyl acetate Toluene Benzene THF Dichloromethane Diethyl ether Acetone Acetonitrile

Hildebrand Solubility Parameter 25.9 ± 1.0 MPa^{1/2}

[Open Solvent Prediction Based on the Hildebrand Parameter](#)

Permeability Properties

Gas	He	H ₂	CO ₂	N ₂	O ₂	CH ₄
Permeability (Barrer)	2.8	2.1	1.4	3.7e-1	7.5e-1	2.6e-1
log(Permeability)	0.5 ± 2.3	0.3 ± 2.3	0.1 ± 2.3	-0.4 ± 2.3	-0.1 ± 2.3	-0.6 ± 2.3
Gas1/Gas2	O ₂ /N ₂	CO ₂ /CH ₄	CO ₂ /N ₂	H ₂ /CH ₄	H ₂ /CO ₂	N ₂ /CH ₄
Selectivity	2.0	5.2	3.8	8.0	1.5	1.4

Thermal Properties

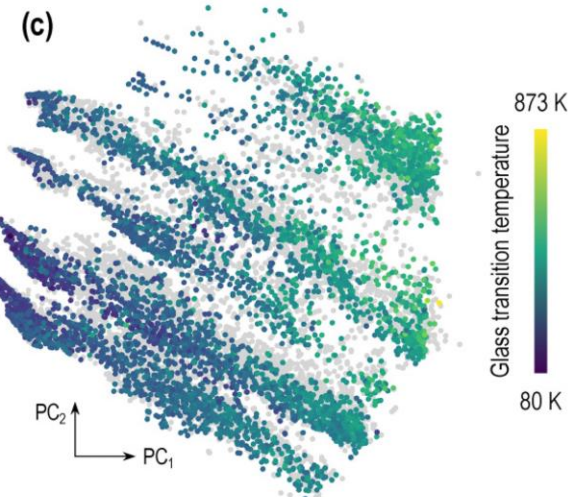
Glass Transition Temperature	551 ± 30 K
Melting Temperature	848 ± 43 K
Thermal Decomposition Temperature	781 ± 48 K

Physical & Thermodynamic Properties

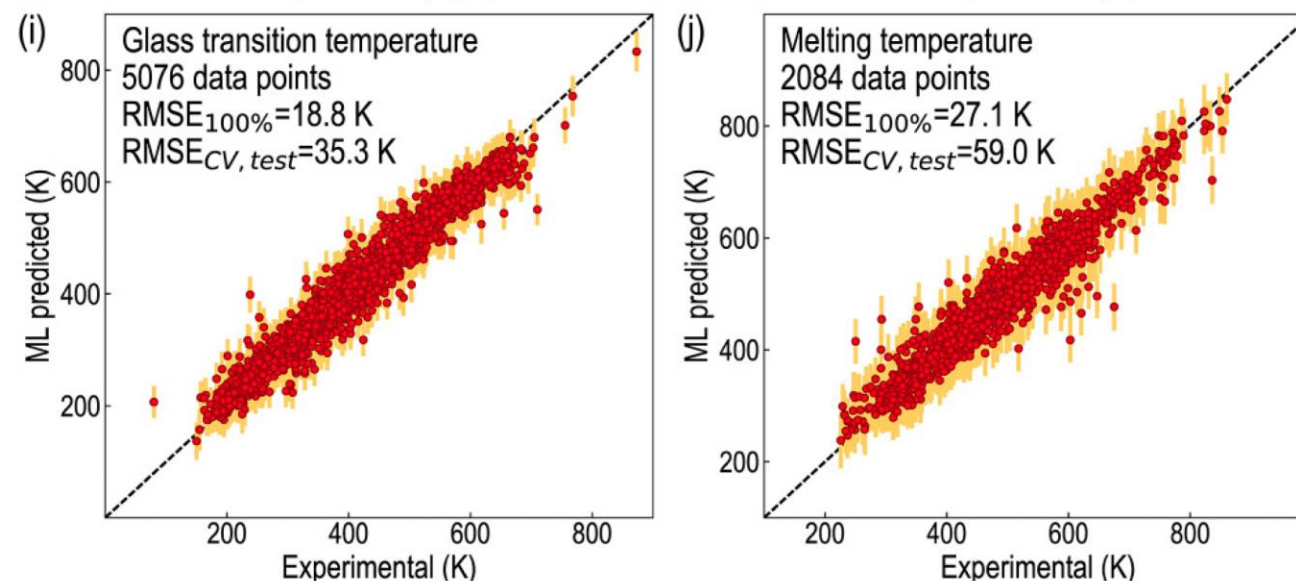
Density	1.3 g/cc
Atomization Energy	-6.4 eV/atom
Specific Heat	1 J/gK
Fractional Free Volume	< 0.1

Other Properties

Tendency to Crystallize	71 ± 21 %
Limiting Oxygen Index	44 ± 10 %



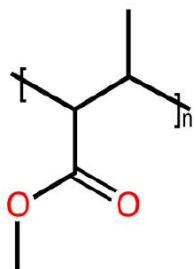
J. Appl. Phys. 128, 171104 (2020)



Success Story of Molecular-level fingerprints

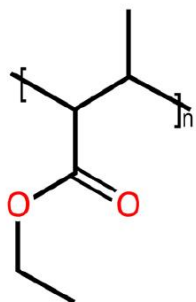
Poly(methyl methacrylate)
C(C)C(C(=O)OC)

$T_{g, \text{Predicted}} = 361 \pm 26 \text{ K}$
 $T_{g, \text{Experimental}} = 373 \text{ K} - 393 \text{ K}$



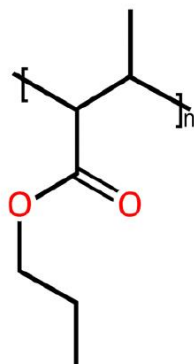
Poly(ethyl methacrylate)
C(C)C(C(=O)OCC)

$T_{g, \text{Predicted}} = 330 \pm 25 \text{ K}$
 $T_{g, \text{Experimental}} = 338 \text{ K}$



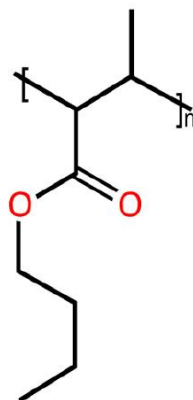
Poly(propyl methacrylate)
C(C)C(C(=O)OCCC)

$T_{g, \text{Predicted}} = 307 \pm 24 \text{ K}$
 $T_{g, \text{Experimental}} = 308 \text{ K}$



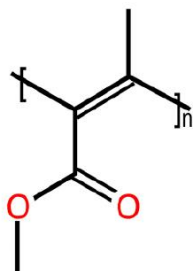
Poly(butyl methacrylate)
C(C)C(C(=O)OCCCC)

$T_{g, \text{Predicted}} = 289 \pm 24 \text{ K}$
 $T_{g, \text{Experimental}} = 293 \text{ K}$

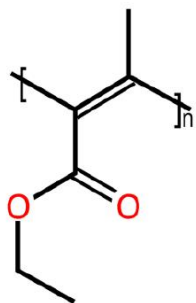


Possible limitations of the
molecular fingerprint?

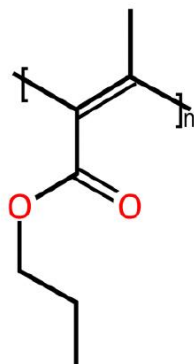
C(C)=C(C(=O)OC)
 $T_{g, \text{Predicted}} = 501 \pm 55 \text{ K}$



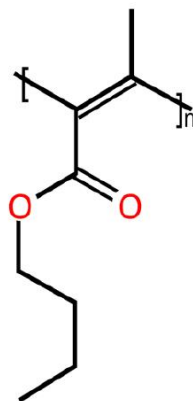
C(C)=C(C(=O)OCC)
 $T_{g, \text{Predicted}} = 474 \pm 46 \text{ K}$



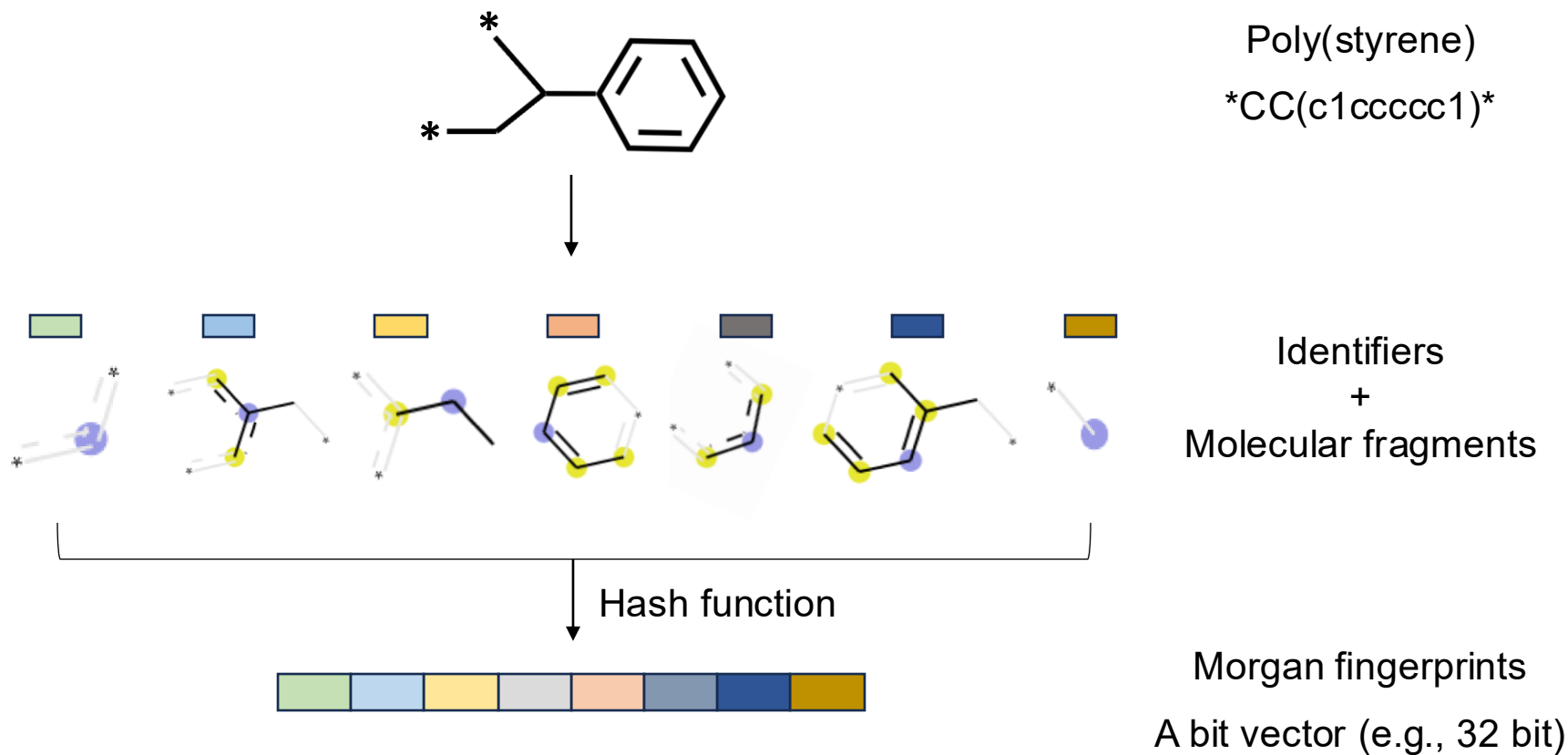
C(C)=C(C(=O)OCCC)
 $T_{g, \text{Predicted}} = 451 \pm 42 \text{ K}$



C(C)=C(C(=O)OCCCC)
 $T_{g, \text{Predicted}} = 426 \pm 40 \text{ K}$



Another Popular Molecular-level fingerprint



- Size of the molecular fragments decided by the radius parameter
- Another variation with fragment frequency
- Similar to ECFP fingerprints

Open-source package:
RDKit(<https://www.rdkit.org/docs/index.html>)

Some other Molecular-level fingerprints

Open sources: RDKit (<https://www.rdkit.org/docs/index.html>)

Fragment name	Definition
NHOHCount	Number of NHs and OHs
NOCOUNT	Number of Nitrogen and Oxygen atoms
NumHAcceptors	Number of Hydrogen Bond Acceptors
NumHDonors	Number of Hydrogen Bond Donors
fr-Al-COO	Number of aliphatic carboxylic acids
fr-Al-OH	Number of aliphatic hydroxyl groups
fr-Al-OH-noTert	Number of aliphatic hydroxyl groups excluding tert-OH
fr-ArN	Number of N functional groups attached to aromatics
fr-Ar-COO	Number of Aromatic carboxylic acids
fr-Ar-N	Number of aromatic nitrogens
fr-Ar-NH	Number of aromatic amines
fr-Ar-OH	Number of aromatic hydroxyl groups
fr-COO	Number of carboxylic acids

Some other Molecular-level fingerprints

Open sources: RDKit (<https://www.rdkit.org/docs/index.html>)

RDKit Descriptors	Paper
MolWt, HeavyAtomMolWt, NumRadicalElectrons, NumValenceElectrons, HeavyAtomCount, NumHeteroatoms, NumRotatableBonds, RingCount	Self-explanatory; the implementation can be found in the open source RDKit version 2012.12.1 descriptor module
Chi0v, Chi1v, Chi2v, Chi3v, Chi4v, ChiNv, HallKierAlpha, Kappa1, Kappa2, Kappa3	Rev. Comp. Chem. vol 2, 367-422, (1991)
Chi0n, Chi1n, Chi2n, Chi3n, Chi4n, ChiNn	Similar to Hall Kier ChiXv, but uses nVal instead of valence
BalabanJ	Chem. Phys. Lett. vol 89, 399-404, (1982)
BertzCT	J. Am. Chem. Soc., vol 103, 3599-601 (1981)
Ipc	J. Chem. Phys., vol 67, 4517-33 (1977)
LabuteASA	J. Mol. Graph. Mod., vol 18, 464-77 (2000)
PEOE-VSA1 – PEOE-VSA14	
SMR-VSA1 – SMR-VSA10	
SlogP-VSA1 – SlogP-VSA12	
TPSA	J. Med. Chem., vol 43, 3714-7, (2000)
MolLogP, MolMR	J. Chem. Inform. Comput. Sci., vol 39, 868-73 (1999)
EState-VSA1 – EState-VSA11	MOE-type descriptors using electrotopological state indices and surface area contributions developed at RD
VSA-EState1 – VSA-EState10	
	from J. Chem. Inform. Comput. Sci., vol 31, 76-81 (1991)

- Collection of several type of molecular descriptors
- Some molecular fragment-based, some graph-based.

Some other Molecular-level fingerprints

Open sources: RDKit (<https://www.rdkit.org/docs/index.html>)

No. of non-hydrogen
neighbors



No. of valence e- shared
with non-hydrogen
neighbors



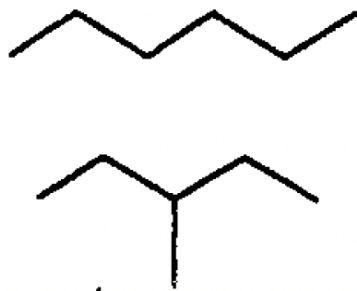
For each vertex (atom)

$${}^0c_s = (\delta_s)^{-1/2} \quad \text{and} \quad {}^0c_s^v = (\delta_s^v)^{-1/2}$$

For graph (molecule)

$${}^0\chi = \sum {}^0c_s \quad \text{and} \quad {}^0\chi^v = \sum {}^0c_s^v$$

Example



Some other Molecular-level fingerprints

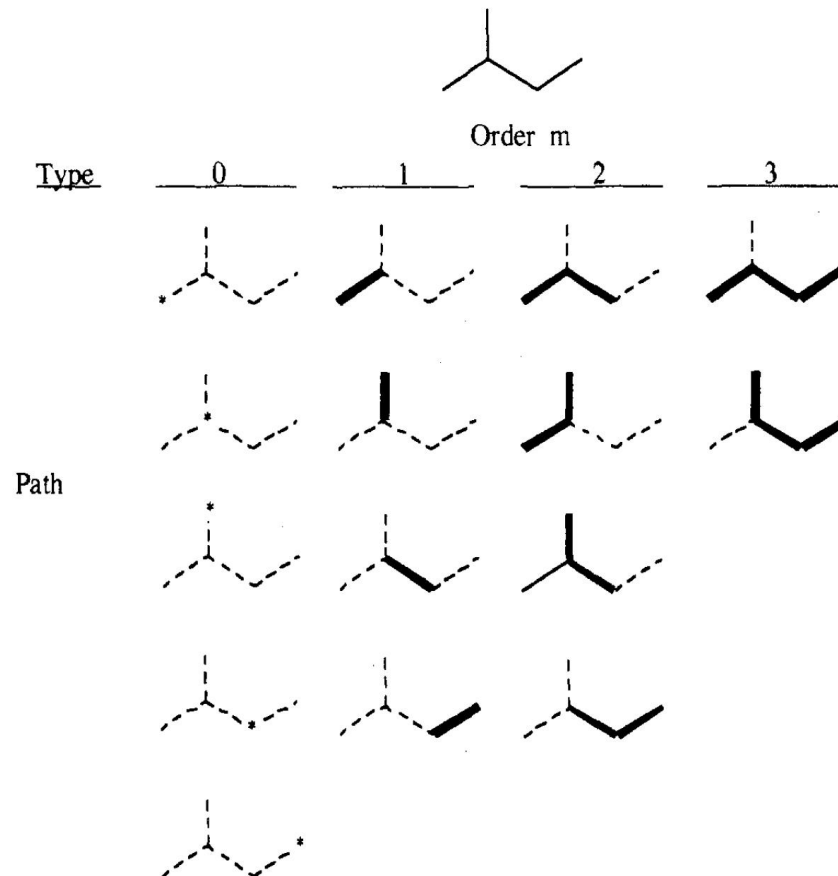
Open sources: RDKit (<https://www.rdkit.org/docs/index.html>)

For each edge (atom) ${}^1c_s = (\delta_i \delta_j)_s^{-1/2}$ and ${}^1c_s^v = (\delta_i^v \delta_j^v)_s^{-1/2}$

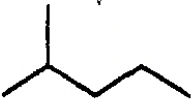
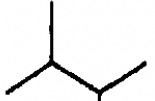
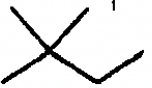
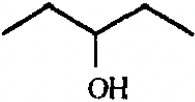
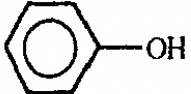
For graph (molecule) ${}^1\chi = \sum {}^1c_s$ and ${}^1\chi^v = \sum {}^1c_s^v$

For each edge (atom) ${}^m c_s = \Pi(\delta_i)_s^{-1/2}$ and ${}^m c_s^v = \Pi(\delta_i^v)_s^{-1/2}$

For graph (molecule) ${}^m \chi_t = \sum {}^m c_s$ and ${}^m \chi_t^v = \sum {}^m c_s^v$



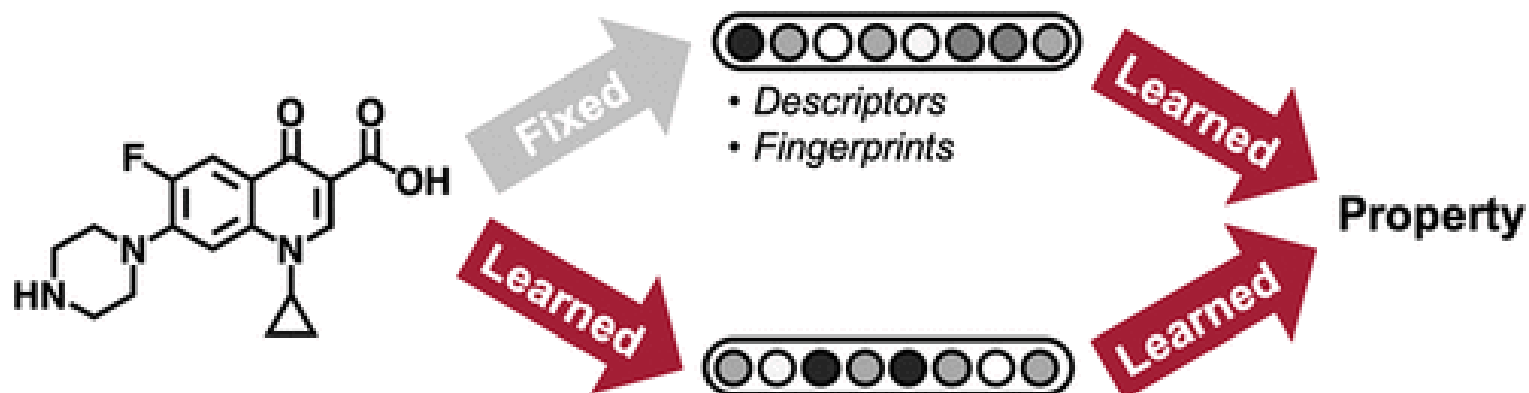
Graph-based Molecular-level fingerprints

Graph	0p	1p	2p	3p	4p	5p	$^0\chi$	$^1\chi$	$^2\chi$	$^3\chi_p$	$^4\chi_p$	$^5\chi_p$
	6	5	4	3	2	1	4.828	2.914	1.707	0.957	0.500	0.250
	6	5	5	4	1	0	4.992	2.808	1.922	1.394	0.289	0.000
	6	5	5	3	2	0	4.992	2.770	2.183	0.866	0.577	0.000
	6	5	6	4	0	0	5.155	2.643	2.488	1.333	0.000	0.000
	6	5	7	3	0	0	5.207	2.561	2.914	1.061	0.000	0.000
	6	5	4	3	2	1	4.276	2.952	1.431	0.762	0.362	0.112
	6	5	5	4	1	0	4.439	2.489	1.470	0.943	0.289	0.000
	6	5	7	3	0	0	4.654	2.284	2.916	0.865	0.000	0.000
	7	7	8	8	8	8	3.834	2.134	1.336	0.756	0.428	0.029

Can we improve upon Chi scores?

- Ad hoc transformation in definition of δ
- No cross-talking between δ and δ^v
- Other properties beyond count of neighbours, valency such as atomic weight, number, electronegativity, etc.

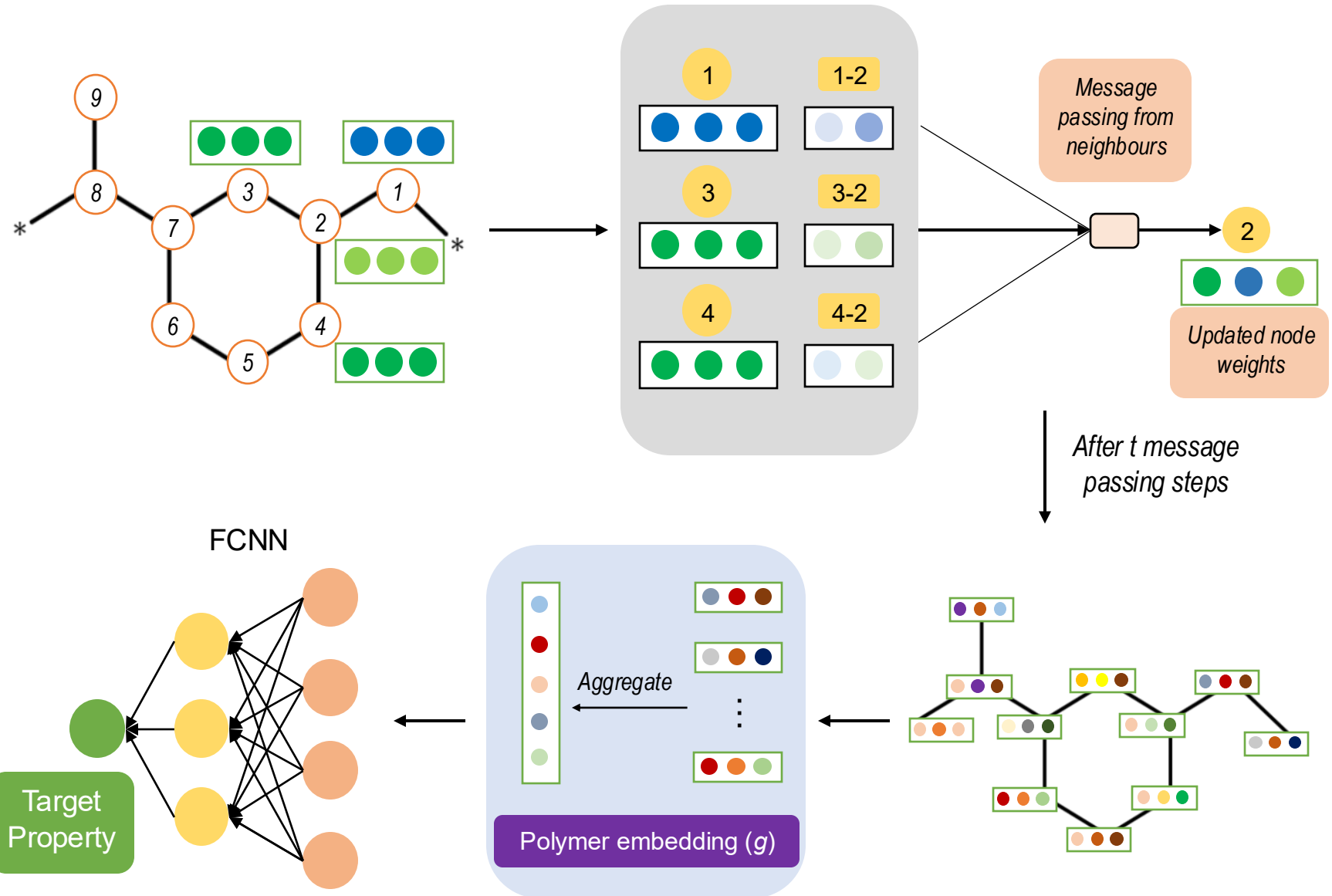
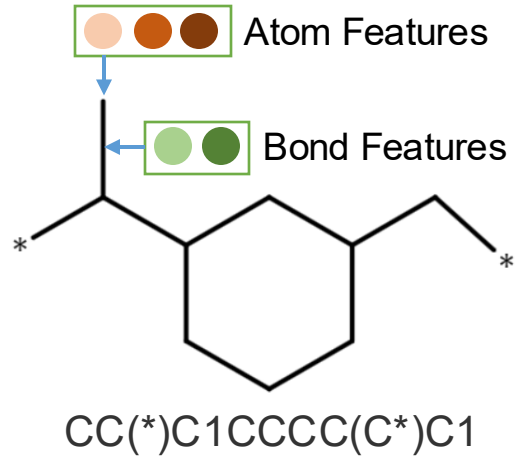
Can we learn Molecule fingerprints



Possible ML methods:

- Graph-based (e.g., graph convolution NN)
- Transformer-based

Graph Convolution NN

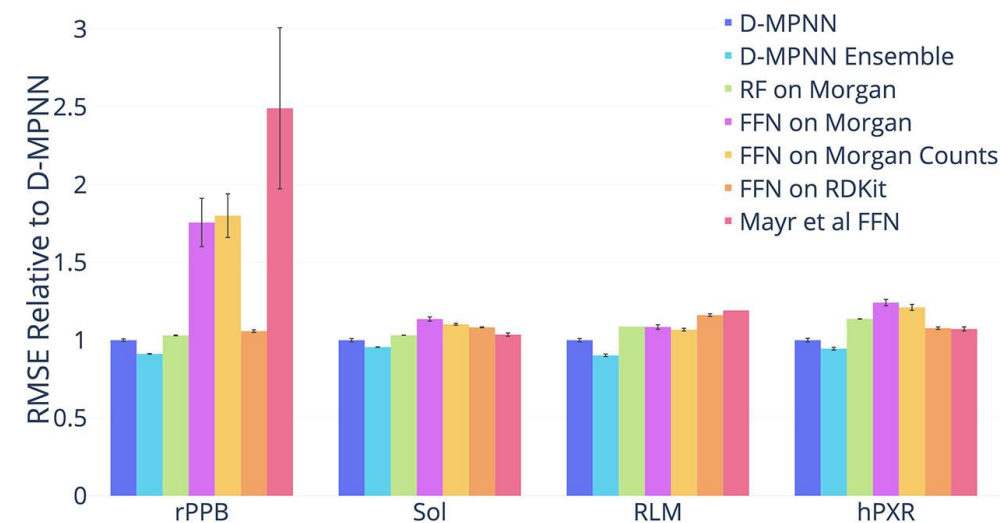


Graph Convolution NN

Datasets

data set	no. of tasks	task type	no. of compounds	metric
QM7	1	regression	6,830	MAE
QM8	12	regression	21,786	MAE
QM9	12	regression	133,885	MAE
ESOL	1	regression	1,128	RMSE
FreeSolv	1	regression	642	RMSE
Lipophilicity	1	regression	4,200	RMSE
PDBbind-F	1	regression	9,880	RMSE
PDBbind-C	1	regression	168	RMSE
PDBbind-R	1	regression	3,040	RMSE
PCBA	128	classification	437,929	PRC-AUC
MUV	17	classification	93,087	PRC-AUC
HIV	1	classification	41,127	ROC-AUC
BACE	1	classification	1,513	ROC-AUC
BBBP	1	classification	2,039	ROC-AUC
Tox21	12	classification	7,831	ROC-AUC
ToxCast	617	classification	8,576	ROC-AUC
SIDER	27	classification	1,427	ROC-AUC
ClinTox	2	classification	1,478	ROC-AUC
ChEMBL	1310	classification	456,331	ROC-AUC

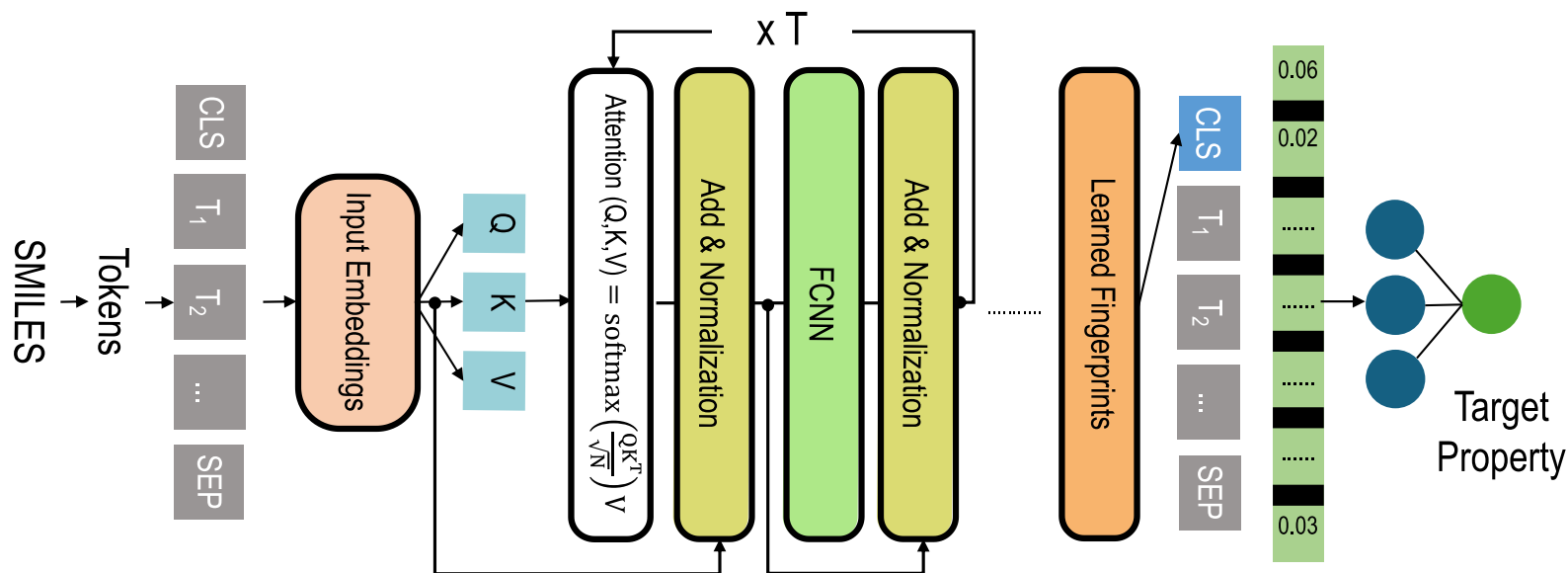
Results



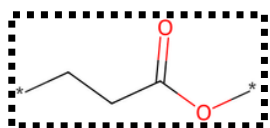
(a) Regression Data Sets (lower = better).

baseline	D-MPNN is <i>better</i>	D-MPNN is the <i>same</i>	D-MPNN is <i>worse</i>	no. of data sets
MoleculeNet (2)	5	3	2	10
Mayr et al. (12)	8	10	1	19
RF on Morgan	14	0	1	15
FFN on Morgan	14	5	0	19
FFN on Morgan Counts	15	4	0	19
FFN on RDKit	8	5	4	19

Transformer-based Models



Parambil, V. et al. Materials Informatics III.
vol 41. Springer, Cham. (2025)
https://doi.org/10.1007/978-3-031-78724-9_5



[*]CCC(=O)O[*]

**Byte-Pair-
Encoding**

[*]	C	C	C	(	=	O	)	O	[*]
0.132	2.49	0.111	0.59	0.63	-0.21	-0.54	0.67	0.22	0.56
0.45	0.34	0.59	0.12	1.76	-0.2	-0.5	-1.3	-2.7	-0.36
.....									
0.009	0.98	-1.54	-0.87	0.01	0.09	0.07	0.68	0.32	2.45

$$\text{Softmax}\left(\frac{Q \times K^T}{\sqrt{N}}\right)$$

	[*]	CC	([*])	(C)	C(=O)	OC	sum
[*]	0.26	0.11	0.13	0.14	0.17	0.01	1
CC	0.12	0.27	0.21	0.12	0.15	0.07	1
([*])	0.14	0.13	0.26	0.09	0.21	0.02	1
(C)	0.21	0.12	0.15	0.23	0.01	0.11	1
C(=O)	0.14	0.15	0.16	0.01	0.43	0.22	1
OC	0.16	0.19	0.12	0.14	0.22	0.32	1

Dimensions:

- Q: (N,E)
- K^T : (E,N)
- QK^T : (N,N)

$$\text{Attention} = V \times \text{Attention} = \text{Attention}$$

Transformer-based Models

Datasets

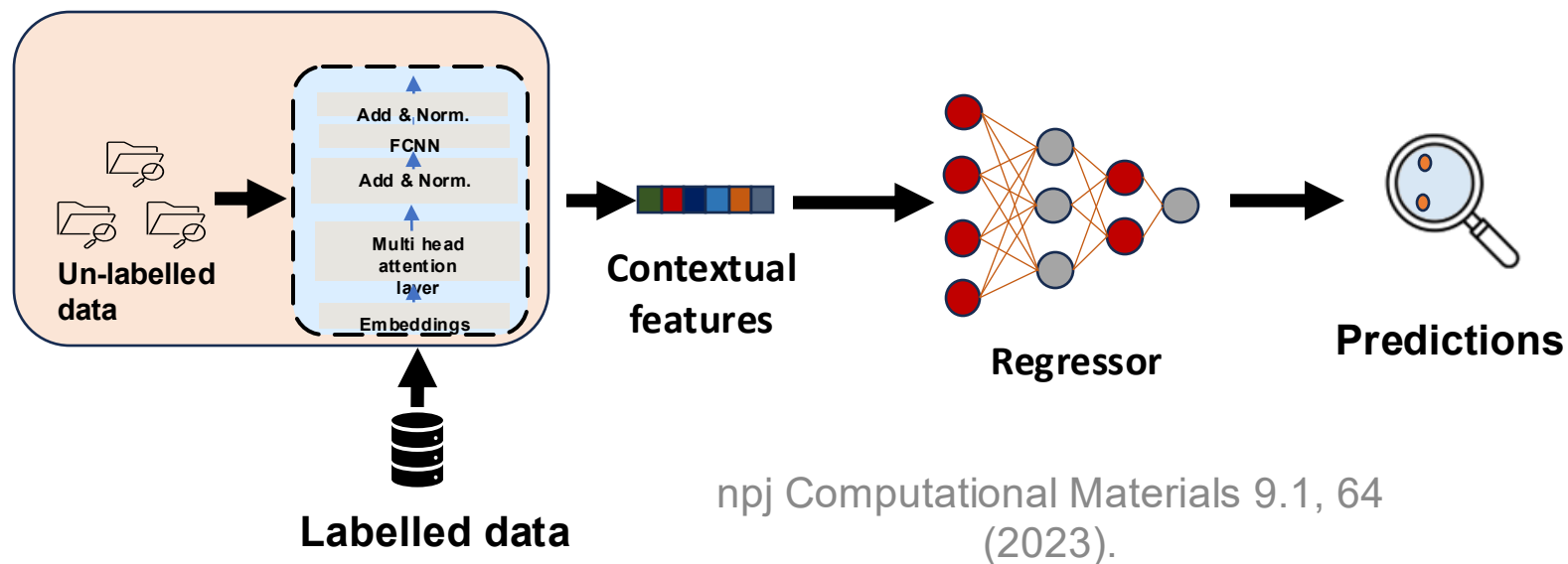
Dataset	Property	# Data	# Augmented train data	# Test data	Data split
PE-I ⁵³	conductivity	9185	34803	146	train-test split by year
PE-II ⁵²	conductivity	271	8864	55	5-fold cross-validation
Egc ⁵⁴	bandgap (chain)	3380	5408	676	5-fold cross-validation
Egb ⁵⁴	bandgap (bulk)	561	6443	113	5-fold cross-validation
Eea ⁵⁴	electron affinity	368	3993	74	5-fold cross-validation
Ei ⁵⁴	ionization energy	370	4000	74	5-fold cross-validation
Xc ⁵⁴	crystallization tendency	432	8837	87	5-fold cross-validation
EPS ⁵⁴	dielectric constant	382	4188	77	5-fold cross-validation
Nc ⁵⁴	refractive index	382	4188	77	5-fold cross-validation
OPV ⁵⁵	power conversion efficiency	1203	4810	241	5-fold cross-validation

Results

Model	Test RMSE (↓)						
	Egc (eV)	Egb (eV)	Eea (eV)	Ei (eV)	Xc (%)	EPS	Nc
Gaussian Process (PG)	0.48	0.55	0.32	0.42	24.42	0.53	0.10
Neural Network (PG)	0.49	0.57	0.32	0.45	20.74	0.54	0.10
Random Forest (ECFP)	0.81	0.88	0.56	0.58	25.61	0.75	0.14
LSTM	0.58	1.94	1.04	0.94	23.67	1.11	0.23
TransPolymer _{unpretrained}	0.63	0.61	0.36	0.46	20.11	0.59	0.10
TransPolymer _{pretrained}	0.44	0.52	0.32	0.39	16.57	0.52	0.10

Xu, C. et al. npj Comput Mater 9, 64 (2023)

LLM Models are good for Property Prediction

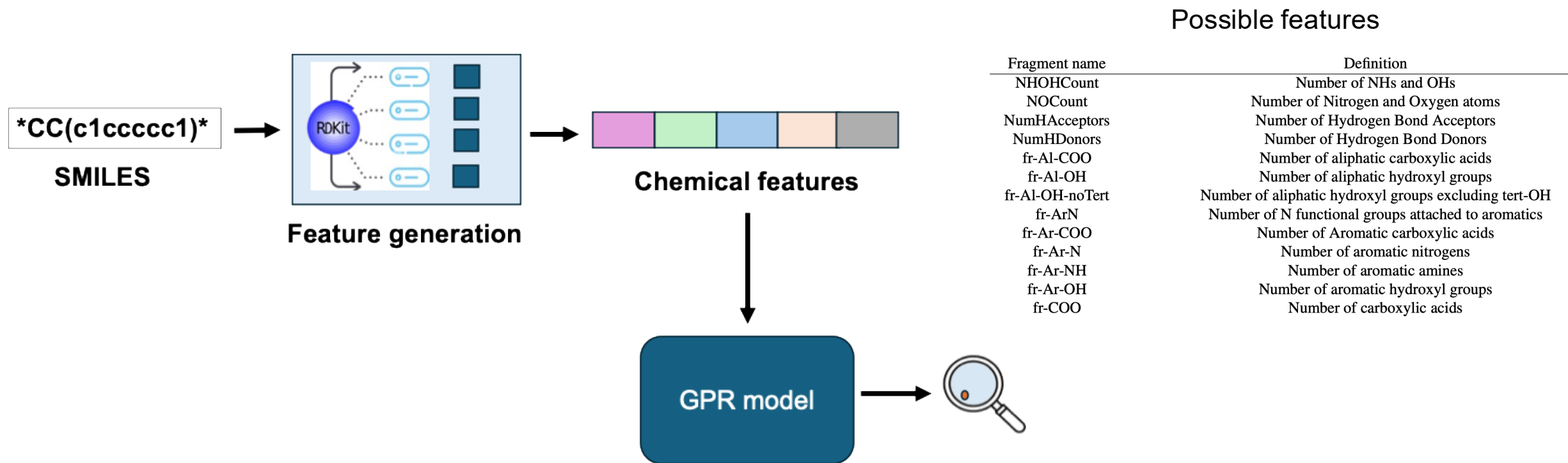


Unlabeled data: SMILES of hypothetical polymers (5M)

Labelled data: Polymers with known property values (~300-3000)

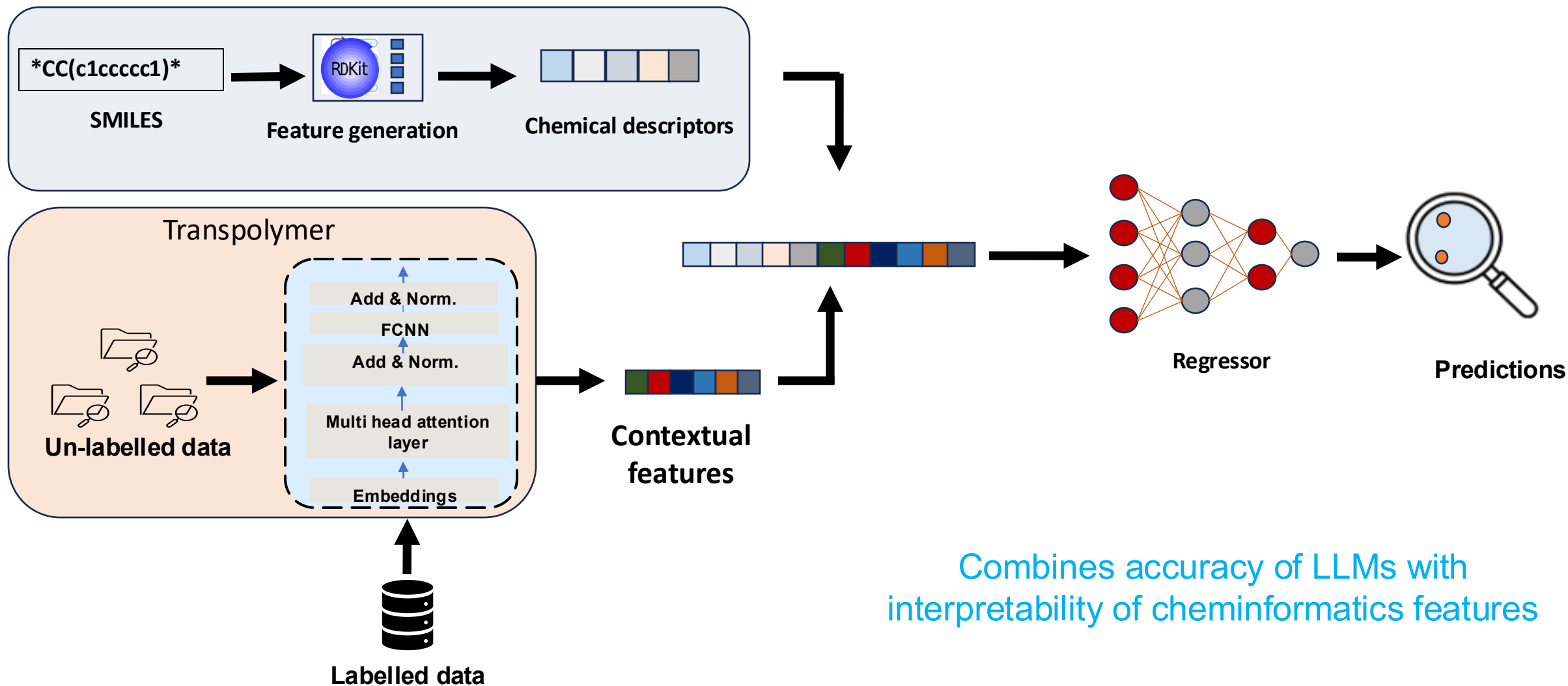
- Perform well for several polymer properties (especially larger datasets)
- Lacks interpretability (chemical reasoning of predictions)

Other (traditional) Models



- Better interpretability (chemical intuition-based)
- Limited performance (slightly lower than LLMs and other graph-based methods)

“TransChem” Architecture



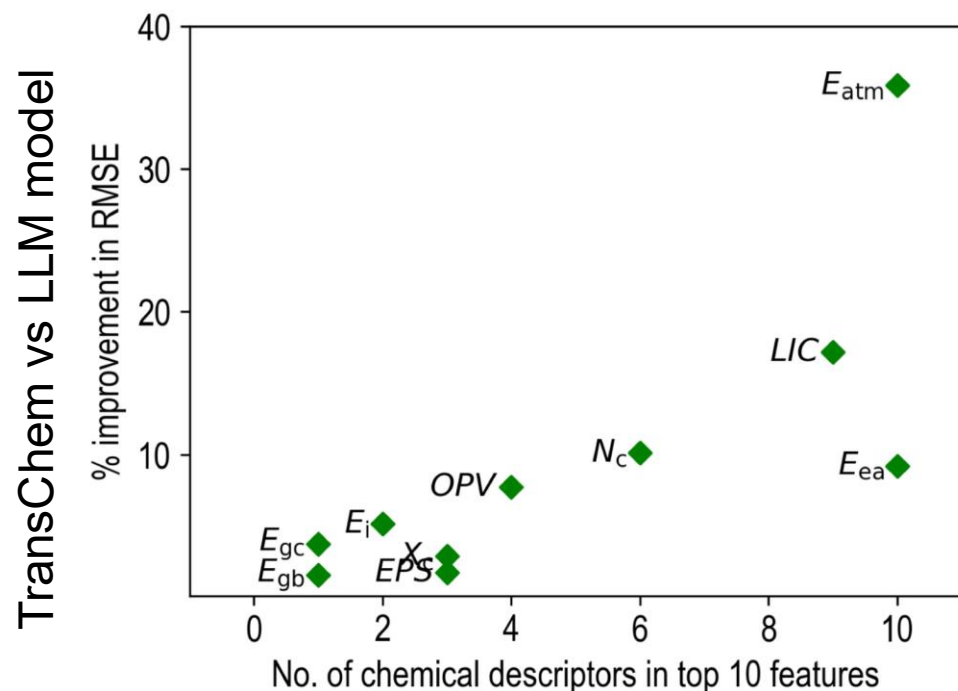
TransChem Model Performance

Dataset	RMSE					R ² score				
	Chem	TP	TC	MT-TP	MT-TC	Chem	TP	TC	MT-TP	MT-TC
E_{ea}	0.393	0.326	0.297	0.300	0.294	0.86	0.902	0.920	0.922	0.925
E_{gb}	0.601	0.518	0.501	0.467	0.508	0.903	0.928	0.933	0.942	0.931
E_{gc}	0.631	0.457	0.439	0.508	0.510	0.834	0.913	0.920	0.888	0.891
E_i	0.452	0.396	0.377	0.399	0.373	0.785	0.830	0.841	0.823	0.837
X_c	17.010	16.782	16.203	–	–	0.483	0.485	0.521	–	–
OPV	2.141	1.923	1.775	–	–	0.161	0.322	0.423	–	–
EPS	0.551	0.531	0.512	0.541	0.543	0.744	0.756	0.778	0.750	0.754
N_c	0.138	0.099	0.089	0.097	0.095	0.663	0.814	0.830	0.824	0.835
LiC	0.721	0.827	0.685	–	–	0.573	0.327	0.623	–	–
E_{atm}	0.039	0.053	0.034	–	–	0.98	0.974	0.989	–	–

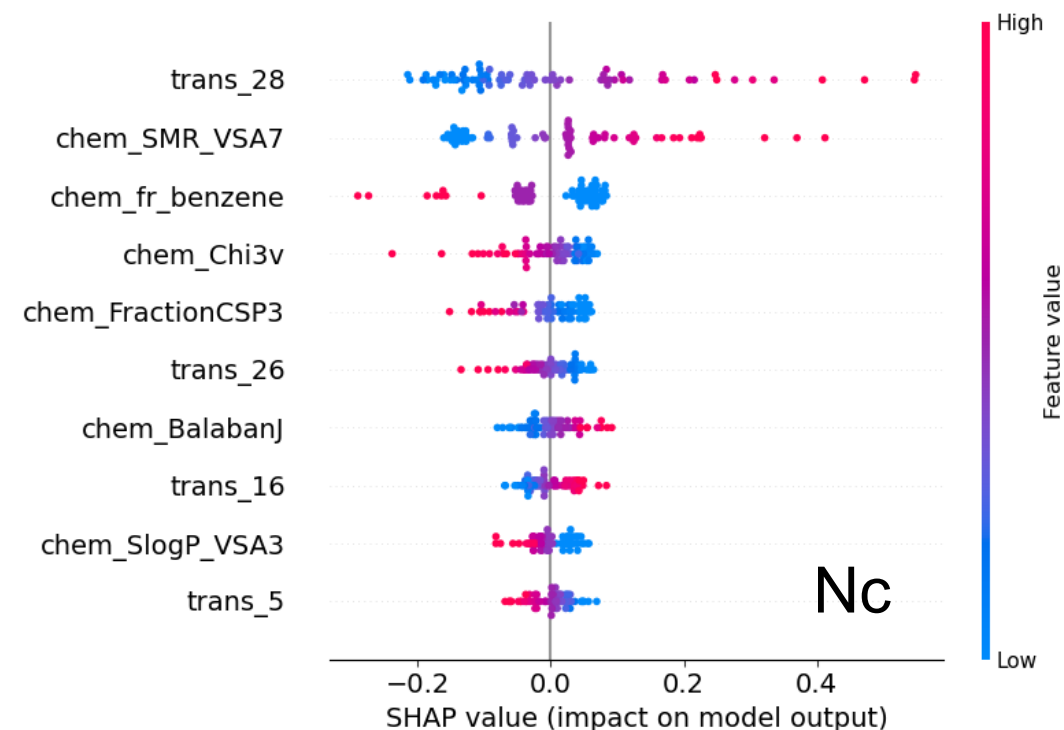
- Compared performance over 10 diverse polymer datasets
- TransChem consistently performs better
- No consistent performance improvement with multi-task variants

Chem: Cheminformatics; TP: TransPolymer; TC: TransChem

Benefits of TransChem



- Larger the number of high importance cheminformatics features, higher the accuracy gain
- Cheminformatics features responsible for improved TransChem performance



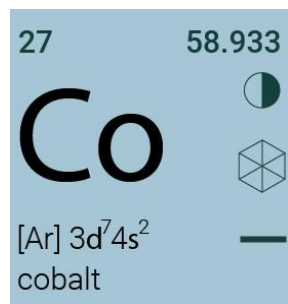
- **trans_28**: transformer feature (not interpretable)
- **chem_SMR_VSA7**: Van Der Waal's surface area
- **chem_fr_benzene**: Number of fraction of benzene rings
- **chem_Chi3v**: Atomic connectivity

TransChem improves accuracy and explainability

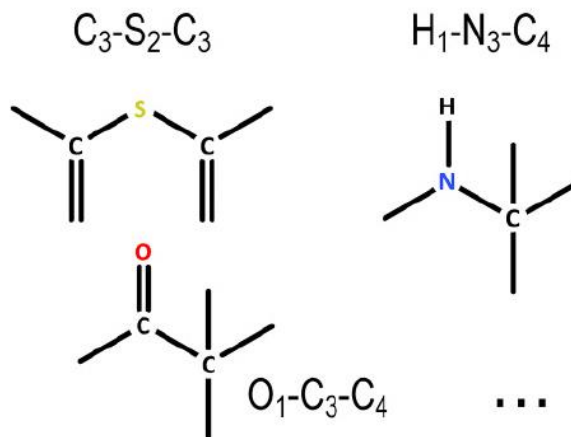
Broad Categories of Fingerprints

- High level (only high-level details)
- Molecular fragment level (medium level details)
- Atomic level (fine details)
- Miscellaneous (driven by ML algorithms)

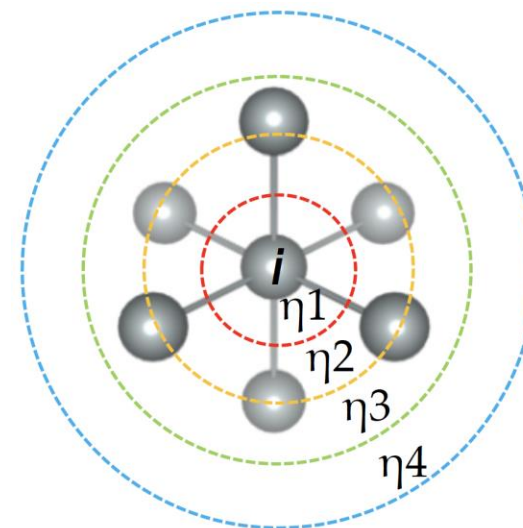
High level



Molecular fragment level



Atomic level



Getting Started: Course Work

- Formal courses:
 - Linear algebra
 - Probability theory
 - Some foundational course in programming (python is highly recommended)
 - Some foundational course on AI/ML
 - A Deep learning course is also advised

Getting Started: Books/Articles

Sobester, András, Alexander Forrester, and Andy Keane. *Engineering design via surrogate modelling: a practical guide*. John Wiley & Sons, 2008, ISBN: 9780470060681.

Notes: Chapter 2,3 (Active learning), Good for general ML knowledge

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition, Aurélien Géron, O'Reilly Media, Inc., 2019 (ISBN: 9781492032649)

Notes: Beginner level. Hands-on style learning. Code available. Shallow in concepts but good to get started with making different types of ML models.

Mueller, Tim, Aaron Gilad Kusne, and Rampi Ramprasad. *Machine learning in materials science: Recent progress and emerging applications*. *Reviews in computational chemistry* 29 (2016): 186-273.

Notes: Medium-level review article. Good for people with research inclination.

Gaussian processes for machine learning. MIT Press, 2005, ISBN: 9780262182539.

Notes: Go to book for Gaussian process regression. (Good for probabilistic ML)

Probability - Random Variables and Stochastic Processes (4th Edition) by Athanasios Papoulis, S Pillai

Notes: Excellent book on Probability theory

Andrew White, *Deep Learning for Molecules and Materials*, <https://whitead.github.io/dmol-book/intro.html>

Notes: Beginner level. Hands-on style learning. Codes available. Good for students into deep learning or molecule design.

Ian J. Goodfellow, Yoshua Bengio and Aaron Courville. *Deep Learning*, MIT Press, 2016, ISBN-13: 978-0262035613

Notes: Go to book for deep learning. A bit outdated now.

Artificial Intelligence for Science, Alok Choudhary (Northwestern University, USA), Geoffrey Fox (University of Virginia, USA), and Tony Hey (Rutherford Appleton Laboratory, UK), Non-Series Books. April 2023

Materials Informatics: Methods, Tools, and Applications, Olexandr Isayev (Editor), Alexander Tropsha (Editor), Stefano Curtarolo (Editor), ISBN: 978-3-527-34121-4, December 2019

Krishna Rajan; *Informatics for Materials Science and Engineering*. Butterworth-Heinemann, 2013, ISBN 978-0-12-394399-6. 3. Rasmussen, C. E., & Williams, C. K. I.

Hands On – ML

Step 1: Scan the QR code and go to the link

<https://drive.google.com/drive/folders/1QG5WtQc6YpWJvHFChqRwIXp-7hi7N8ms?usp=sharing>

Step 2: Download file “Tutorial_Polymer_fingerprint_extended.ipynb” from the folder

Step 3: Upload the file in your Google Drive account

Step 4: Right-click and choose “Open With” -> “Google Colaboratory”

NOTE: If “Google colaboratory” widget is not available, you will have to add “colab” app to your Google workspace

